MILITARY SPECIFICATION SHEET

ELECTRON TUBE, MAGNETRON

TYPE 2J51A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Pulsed, mechanically tunable frequency 8,500 to 9,600 MHz, 45 kw peak power output, integral magnet, air cooled

ABSOLUTE RATINGS:

Parameter:	Ef	tpc	Du	tk	Pi	ib	VSWR	T(anode)	Tuner torque	Alt
Unit:	V	µs	---	sec	W	a	---	°C	in.-lb	ft
Maximum:	7.0	3.4	0.0011	---	230	15.5	1.5	150	2.5	10,000
Minimum:	---	0.1	---	120	---	12.5	---	-55	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1	Marking:	See note 1 and figure 1
Mounting position:	Any	Magnet isolation:	See note 3 and figure 2
Mounting support:	Mounting flange (see figure 1)	Weight:	5 pounds (approx)
Output coupling:	See note 2 and figure 1		

TEST CONDITIONS:

Parameter:	Ef	tpc	trv	Du	Ib	VSWR	F
Unit:	V	µs	µs (max)	---	mAdc	--- (max)	MHz
Test condition 1:	0	1.0 ±0.1	0.07	0.001	14	1.10:1	F1, F3, F5
Test condition 2:	5.0	0.11 ±0.01	0.07	0.00033	4.7	1.10:1	F1, F3, F5
Test condition 3:	0	3.50 ±0.1	0.12	0.0011	15.5	1.10:1	F1, F3, F5

Frequency (See note 1)		
F	MHz	±
F1	8,500	10
F2	8,600	10
F3	9,000	10
F4	9,375	13
F5	9,600	20

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
4028	Barometric pressure, reduced	-	1	F5; pressure = 380 mmHg	---	---	---	---
4037	Temperature coefficient	Fig 1 (t) and fig 2	1	F3; Ib = 10 mAdc; T(anode) = 70° to 100°C; magnetic field, 4 shunts	ΔF/ΔT	---	0.25	MHz/°C
1047	Low-temperature operation	-	3	F5; Ib = 17.5 mAdc; VSWR = 1.5:1	---	---	---	---
1143	Forced cooling	12	1	Pi-Po = 150 W; TA = 50°C (max)	T	---	TA + 50	°C
4306	Pulse voltage	Fig 2	1	F3; magnetic field, 4 shunts	epy	9	11	kv
1032	Vibration, mechanical	13,14	-	No voltages applied	---	---	---	---
1042	Shock, specified pulse	14	-	Test condition A, except 15G; no voltages applied	---	---	---	---
4223	Operating torque or force	-	-	TA = -55° to +125°C	---	---	2.0	in.-lb
4265	Direct-interelectrode capacitance	-	-		C	4.0	8.0	pF
4225	Mechanical tuning fatigue	15	-		---	2,500	---	Cycles
---	Voltage tuning	16	1		Δepy	0	+1.5	kv
---	Tuning characteristics	11	1		---	---	---	---
	Quality conformance inspection, part 1	4	-					
4003	Pressurizing	-	-	40 psia	---	---	---	---
4299	Heater current, nonoperating	-	-	Ef = 6.3 V; tk = 120 (max)	If	0.90	1.10	A
4306	Pulse voltage	-	3	F3	epy	13	15	kv
4250	Power output (1)	-	3		Po	44	---	W
4315	Pulse stability (1)	5	3	Ib = 17.5 mAdc; VSWR = 1.5:1	MP	---	1.0	%
	Quality conformance inspection, part 2							
4308	RF bandwidth	6	3	VSWR = 1.5:1; Ib = 13.5 and 17.0 mAdc	BW	---	3.5/tpc	MHz
		6	1	VSWR = 1.5:1; Ib = 12.5 and 15.5 mAdc	BW	---	2.5/tpc	MHz
		6	2	Ib = 4.1 and 5.1 mAdc	BW	---	2.20/tpc	MHz
4308	Minor lobe ratio	6	1	Ib = 12.5 and 15.5 mAdc	SL	6	---	dB
		6	2	VSWR = 1.5:1; Ib = 4.1 and 5.1 mAdc	SL	8	---	dB

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued							
4250	Power output (2)	7 -	1 2	F1, F2, F3, F4, and F5 F3	Po Po	40 13	--- ---	W W
4223	Mechanical tuning range	-	1		F	9,500	9,600	MHz
4223	Operating torque or force	-	-	TA = 25° ±5°C	---	---	10	in.-oz
4223	Resettability	-	1	F4	±F	---	10	MHz
4315	Pulse stability (2)	5	2	VSWR = 1.5:1; Ib = 5.3 mAdc	MP	---	1.0	%
4310	Frequency pulling figure	8	1		ΔF	---	18	MHz
	Quality conformance inspection, part 3							
4551	Intermittent life	9	-	Group D; VSWR = 1.5:1 (min); cycled through λg every 30 minutes (approx)	--- t	250 500	--- ---	Cycles hrs
---	Intermittent life-test end points:							
4223	Resettability	10	1	F4	ΔF	---	15	MHz
4250	Power output (1)	10	3		Po	36	---	W
4315	Pulse stability (2)	5,10	2	VSWR = 1.5:1; Ib = 5.3 mAdc	MP	---	2	%
4315	Pulse stability (1)	5,10	3	VSWR = 1.5:1; Ib = 17.5 mAdc	MP	---	2	%
4308	RF bandwidth	6	3	Ib = 13.5 and 17.0 mAdc	BW	---	4.2/tpc	MHz

NOTES:

1. In addition to regular markings, the tuner dial readings for the following frequencies, with the exception of 9,000 MHz shall be stamped on the tube:

Frequency	Marking
8,500 ±10 MHz	F1
8,600 ±10 MHz	F2
9,000 ±10 MHz	F3
9,375 ±13 MHz	F4
9,600 ±20 MHz	F5

These markings shall apply after thermal equilibrium under the conditions of test condition (3) at an anode temperature of 80° ±10°C. These frequency ranges shall be obtained by rotating the tuning dial gear in a continuously clockwise direction.

2. The TUT may be coupled directly to a M85/1-073 waveguide with a M3922/59-007 choke flange. However, for a minimum VSWR at this coupling, it is recommended that a specially designed choke flange be used which mates with the angular M85/1-073 output of the tube to provide a straight section of M85/1-073 waveguide through the coupling network.

3. In handling and mounting the tube, care shall be exercised to prevent demagnetization. Ferromagnetic materials or energized magnets shall not be brought within 2 inches (50.8 mm) of the tube.

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NOTES: -Continued

4. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.

5. The missing pulses (MP) shall be counted during the last 3 minutes of a test interval not to exceed 6 minutes. A missing pulse is defined as one whose energy within a ±1 percent frequency range of the normal test frequency is 70 percent or less than that of a normal pulse.

6. The rf bandwidth and minor lobes shall be within the limits specified when a VSWR of 1.5:1 is introduced in the load at a distance no greater than 1/2 meter from the TUT coupling flange, the phase being adjusted at a point to be determined by the tube manufacturer for the worst spectrum.

7. This test shall be conducted at a sufficient number of frequencies to insure that power output is above the minimum value throughout the specified tuning range.

8. The pulling measurement shall be made at the frequency of maximum power output. This frequency to be determined by the tube manufacturer.

9. The life-test cycle shall be as specified below:

Test condition	Ib	Ef	Duration
Warm up	0	6.3	2 minutes
2	4.7	5.0	60 minutes
3	15.5	0	60 minutes
Off	0	0	58 minutes

Starting at F1, the frequency will be increased in 100 MHz increments every 10 hours.

10. The TUT shall pass all applicable quality conformance inspection, parts 1 and 2, tests at the end of the specified life, with the condition that the criteria for acceptance be modified in accordance with the life-test end points.

11. With the tuning dial adjusted for a frequency of 9,000 MHz, an operating frequency of 9,600 MHz shall be obtained by rotating the tuning shaft 64 ±12 turns such that the large dial gear moves counterclockwise. A frequency of 8,500 MHz shall be attained by turning the shaft 47 ±6 turns in the opposite direction from the 9,000 MHz setting.

12. With an airflow at standard atmospheric pressure of 25 cfm directed at the cooling fins from an orifice of 1-11/64 inches (29.77 mm) and 1-23/64 inches (34.53 mm) the rise above ambient specified shall not be exceeded. The anode temperature shall be measured at that point indicated on figure 1. The ambient temperature shall be approximately 50°C. The orifice shall be located 1/4 inch (6.35 mm) from the cooling fins.

13. The TUT shall be vibrated at a frequency of 50 to 500 to 50 Hz during a 5-minute interval for each plane. The sinusoidal displacement shall be adjusted to maintain acceleration at 5G.

14. At the conclusion of this test, the TUT shall meet the requirements of method 4250, power output (1) and method 4315, pulse stability (1).

15. A cycle consists of two complete excursions each in opposite directions through the tuning range of the TUT. The tuning shaft shall be continuously driven at a speed of approximately 650 rpm. The TUT shall meet end-of-life requirements at completion of this test.

16. The change in voltage required to maintain constant current shall be measured as the frequency is changed from F1, 8,500 MHz to F5, 9,600 MHz.

17. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Air Force - 85

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DSA - ES

User activities:
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Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
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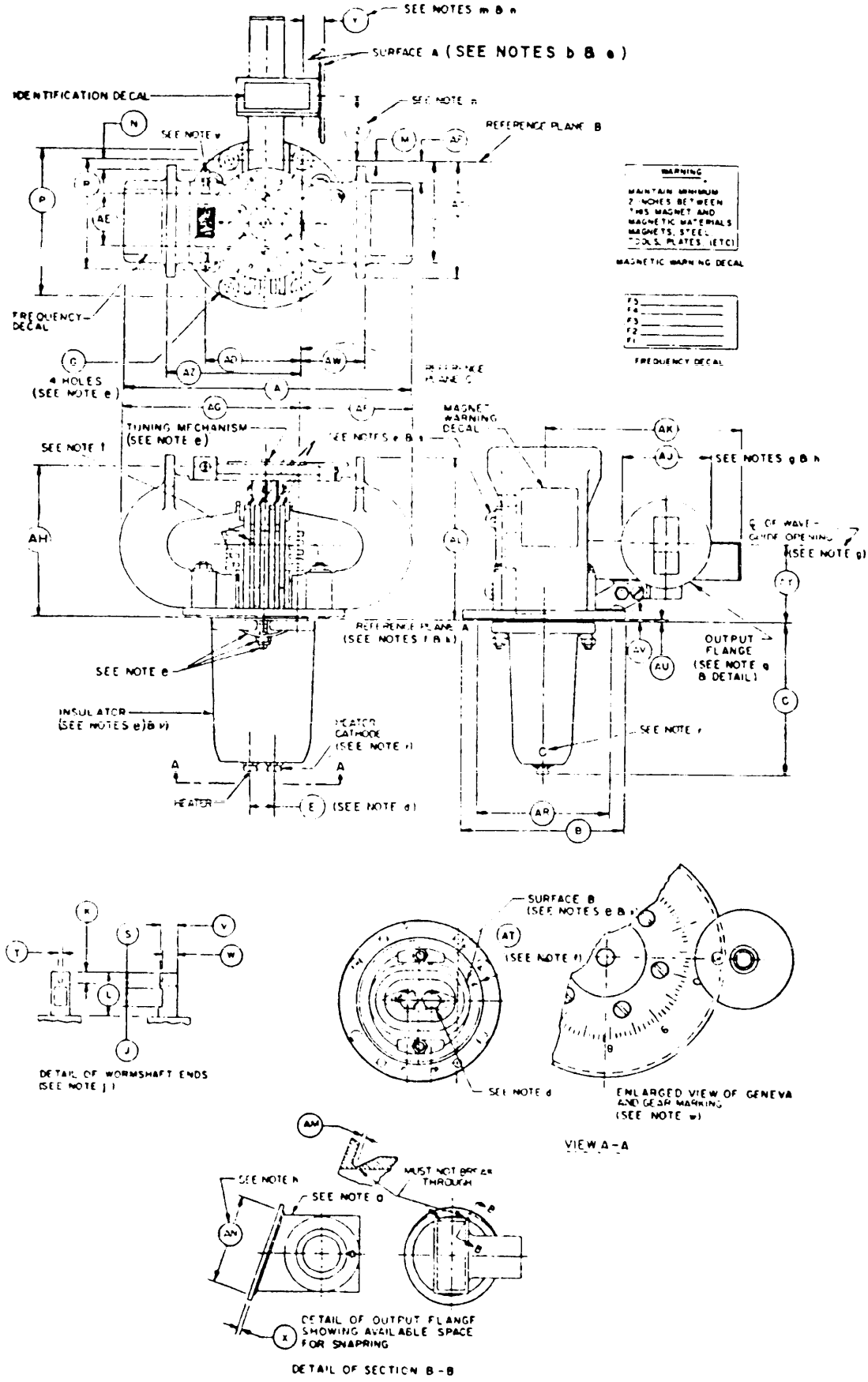


FIGURE 1. Outline drawing of electron tube type 2J51A.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
QUALIFICATION INSPECTION		
W	0.151 (3.84)	0.161 (4.09)
X	0.080 (2.03)	0.090 (2.29)
AU	0.023 (.58)	0.039 (.99)
AV	0.156 (3.96)	
Quality conformance inspection, part 1 (see note c)		
K	0.115 (2.92)	0.135 (3.43)
M	0.042 (1.07)	
N	0.230 (5.84)	0.290 (7.37)
R	2.105 (53.47)	2.165 (54.99)
Y	0.261 (6.63)	0.301 (7.65)
Z	1.173 (29.79)	1.213 (30.81)
AD	1.954 (49.63)	1.984 (50.39)
AE	0.990 (25.15)	1.010 (25.65)
AH	2.834 (71.98)	2.874 (73.00)
AJ	1.743 (44.27)	1.757 (44.63)
AM	0.021 (.53)	0.031 (.79)
AN	1.438 (36.53)	1.446 (36.73)
AX		2.453 (62.31)
AY	1.542 (39.17)	1.582 (40.18)

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A		5.938 (150.83)
B	3.219 (81.76)	3.281 (83.34)
C	2.921 (74.19)	3.047 (77.39)
E	0.490 (12.45)	0.510 (12.95)
G	.190 (4.83)	0.196 (4.98)
J	0.188 (4.78)	0.203 (5.16)
L	0.428 (10.87)	0.448 (11.38)
P	2.869 (72.87)	2.881 (73.18)
S	0.141 (3.58)	0.156 (3.96)
T	0.040 (1.02)	0.045 (1.14)
U		2.188 (55.58)
V	0.187 (4.75)	0.190 (4.83)
AF		2.250 (57.15)
AG		3.688 (93.68)
AK		3.906 (99.21)
AL		3.281 (83.34)
AP	0.308 (7.82)	
AR	2.620 (66.55)	2.630 (66.80)
AT	0.057 (1.45)	0.067 (1.70)
AW		1.344 (34.14)
AZ		2.781 (70.64)

FIGURE 1. Outline drawing of electron tube type 2J51A - Continued.

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NOTES:
a. Section of waveguide M85/1-073.
b. Surface A shall provide hermetic seal with waveguide.
c. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
d. Jacks shall be locking type. Jack holes shall lie within .023 (.58 mm) radius of specified location. Centerlines of holes shall be perpendicular to mounting plate within 3 degrees. Quality conformance inspection, part 2.
e. All metal surfaces shall be painted with heat-resistant, noncorrosive paint except surfaces A and B, tuning mechanism, parts associated with insulator, magnet shunts, four mounting holes G, and area included within .219 (5.56 mm) radius of center of each hole G (qualification inspection). Paint may be omitted from back of output flange.
f. With surface defined by dimension AT and reference plane A resting on a flat surface, a gage .010 (.25 mm) thick and .125 (3.18 mm) wide shall not enter between surfaces at any point. Quality conformance inspection, part 2.
g. Diameter AJ shall be concentric with waveguide opening within .010 (.25 mm).
h. Diameters AJ and AN shall be concentric within .005 (.13 mm).
j. A sleeve .195 (4.95 mm) inside diameter, .406 (10.31 mm) outside diameter, and 1.000 (25.40 mm) long shall pass over ends of shaft to face of worm bracket. Quality conformance inspection, part 2.
k. Any part of assembly which extends below surface defined by reference plane A shall lie within 1.109 (28.17 mm) radius of true center of mounting plate.
m. Tolerances include angular and lateral deviations.
n. Defines relationship between surface A and mounting holes G.
p. Insulator shall be pyrex glass or equal.
q. Protective cover shall be provided for output flange. Waveguide opening shall be kept covered when tube is not in use.
r. Common heater-cathode connection shall be identified by letter C.
s. Four magnet shunts. Number of shunts required shall be in accordance with applicable technical information. Unused shunts may be removed by gripping tabs with pliers and pulling away from tube.
t. Anode temperature shall be measured at this point.
u. Frequency markings F1, F2, F4, and F5 correspond with settings at 8,500, 8,600, 9,375, and 9,600 MHz, respectively.
v. Frequency is increased by tuning this end of wormshaft in counterclockwise direction. Complete frequency range is covered in approximately 125 turns. Tuning mechanism shall operate smoothly over entire mechanical tuning range when subjected to torque of 10 inch-ounces applied to wormshaft. In equipments, not less than 10 inch-ounces nor more than 2.5 inch-pounds shall be applied to the drive shaft. Gear and worm threads shall be free from any obstruction.
w. Number which appears on geneva indicates number of complete revolutions of tuning gear from 0 to 4. With geneva and gear set at 3 and 0, respectively, frequency of tube is 9,000 $\pm$25 MHz under test condition 3.
x. Surface B shall provide hermetic seal with associated mounting surface.

FIGURE 1. Outline drawing of electron tube type 2J51A - Continued.

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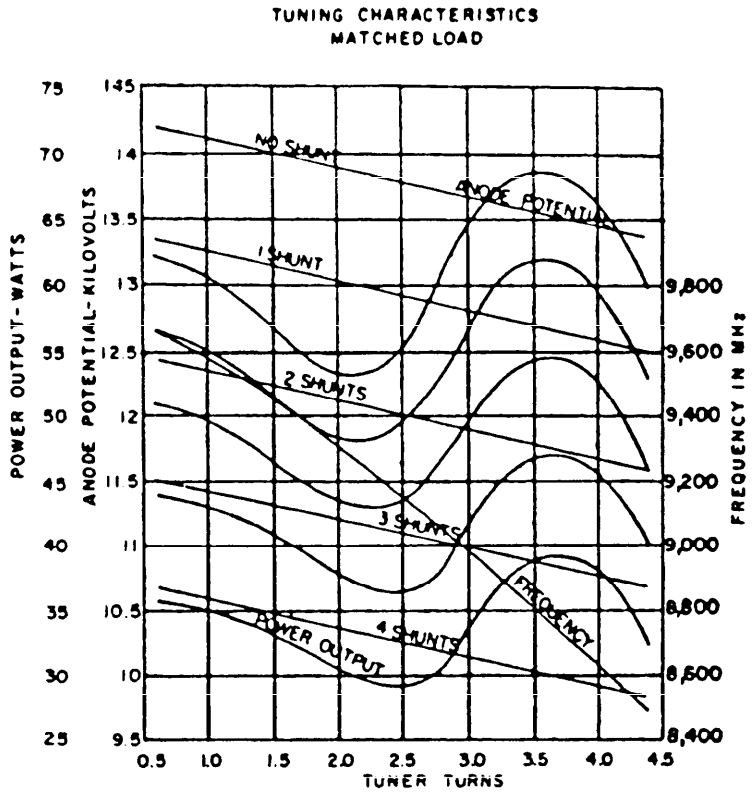


FIGURE 2. Frequency characteristics.

NOTES:

a. Tubes to be supplied with four magnetic shunts. Shunts may be removed as needed to establish various operating points.

b. The chart reveals the manner in which frequency, power output, and anode voltage vary with tuner position for five different gauss levels with a pulse repetition rate of 1,000 pps, a current pulse width of 1.0 microsecond, and a peak anode current of 14 amperes.

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